

Shanilka Haturusinghe

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EDUCATION

University of Kelaniya, Sri Lanka.

March 2020 – October 2024

Bachelor of Science Honours in Computer Science (*First Class*)

Specialization: Artificial Intelligence

GPA: 4.0/4.0

Graduated ranked 1st in the Faculty of Computing and Technology (Class of 2025), across all undergraduate degree programs.

Royal College, Colombo, Sri Lanka.

G.C.E Advanced Level in Mathematics Stream (English Medium)

Results: A in Combined Mathematics, B in Physics, B in Chemistry

PUBLICATIONS

Shanilka Haturusinghe, Tharindu Cyril Weerasooriya, Christopher M Homan, Marcos Zampieri, and Sidath Ravindra Liyanage. 2025. *Subasa – Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala*. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 4: Student Research Workshop)*, pages 260–270, Albuquerque, USA. Association for Computational Linguistics. DOI: 10.18653/v1/2025.naacl-srw.26

<https://aclanthology.org/2025.naacl-srw.26/>

EXPERIENCE - ACADEMIC

University of Kelaniya

August 2025 - Present

Lecturer (Temporary)

- Teach *CSCI 31022 - Machine Learning and Pattern Recognition* & *CSCI 31032 - Theory of Programming Languages* to undergraduate students, covering both theoretical foundations and practical applications.

University of Kelaniya

February 2025 - July 2025

Teaching Assistant & Demonstrator

- Serve as Teaching Assistant for the courses *DSCI 32012 - Advanced Database Applications*, *CTEC 22043 - Object Oriented Programming*, *AINT 32012 - Digital Image Processing and Computer Vision*, and *CSCI 22082 - Object-Oriented Programming — II*.
- Assisted faculty members in updating the course syllabus for better alignment with industry standards.

EXPERIENCE - INDUSTRY

Embla Software Innovation (Pvt) Ltd

October 2023 - April 2024

Software Engineer Intern

- Developed and implemented the "Scrapbook" feature for Embla FT genealogy application with customizable page templates, editable text/photo elements, and comprehensive image/content management capabilities.
- Built full-stack components using C#/.NET Core backend and Angular 9/TypeScript frontend, enhancing existing application architecture to support new scrapbook functionality and user interactions.
- Collaborated with product teams using Agile methodologies, Jira, and Bitbucket for project delivery, utilizing Visual Studio, FastReport Designer, and Figma for development and design tasks.

Xempler Pte. Ltd

September 2022 - May 2023

Intern - Software Developer

- Developed a Sri Lanka License Plate Detection proof of concept using Python and TensorFlow by leveraging YOLO models for robust license plate detection in images. Employed OpenCV for image preprocessing (resizing, filtering, and edge detection) to optimize detection accuracy, and integrated Tesseract OCR to extract alphanumeric text from the detected license plates.
- Designed and implemented a real-time chat feature for the IFAWPCA mobile app, built with React Native. Integrated Firebase for seamless real-time messaging, utilizing Firebase Realtime Database and Firebase Authentication for user management, secure login, and instant message synchronization across devices.

EXPERIENCE - RESEARCH

"Subasa" Research Project with Advisors: Dr. S.R. Liyanage & Dr. Tharindu Cyril Weerasooriya

October 2023 – December 2024

Conducted the *Subasa* project as first author, developing and evaluating low-resource offensive language detection models for Sinhala using novel strategies.

- Developed *Subasa-XLM-R* using a two-stage intermediate pre-finetuning strategy with Masked Rationale Prediction (MRP), where rationale embeddings are fused with token embeddings and 75% of non-special tokens are masked to optimize learning signal for Sinhala context.
- Implemented task-specific instruction fine-tuning for Llama-3 and Mistral models using PEFT with QLoRA, incorporating classification and offensive phrase extraction in a unified prompt framework.
- Conducted comprehensive ablation studies on masking ratios , intermediate task replacement , and direct fine-tuning approaches to validate the effectiveness of the intermediate pre-finetuning strategy.
- Achieved state-of-the-art performance on SOLD benchmark dataset with Subasa-XLM-R (Macro F1: 0.84), outperforming GPT-4o zero-shot performance and existing baselines.
- Open-sourced complete model suite and codebase.

TECHNICAL SKILLS	AI / ML / NLP	Software Engineering
	Python, PyTorch, TensorFlow, Hugging Face Transformers, TRL, Unsloth, LangChain, LangGraph, Sentence-Transformers, ZenML, Google Colab, Google Cloud Compute Engine, AWS SageMaker, Hopsworks, Comet ML, Qdrant	Java, Spring Boot, SQL, PL/SQL, Oracle DB XE 18, C#, .NET Core, Angular 9, React Native, TypeScript, JavaScript, REST Web API, FastAPI, FastReport, Figma, Firebase, Laravel, MongoDB, Git, Docker, Poetry, Selenium, Streamlit

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| PERSONAL PROJECTS | PersonaCraft.AI - Personalized AI Writing Assistant <ul style="list-style-type: none"> • Developed PersonaCraft.AI, an end-to-end ML system for personalized AI writing assistants featuring automated multi-platform web scraping (LinkedIn, Medium, GitHub, custom sources), data extraction pipelines with Selenium, advanced text chunking, and embedding generation using Sentence Transformers. • Engineered a full ML training and MLOps pipeline supporting Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) with TRL and Unsloth, leveraging ZenML for orchestration, MongoDB for data warehousing, Qdrant for retrieval, AWS SageMaker for cloud training, and integrated automated publishing and monitoring through Hugging Face Hub. |
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- Real-Time Fashion Recommendation System**
- Developed an MLOps-driven fashion recommender system using H&M’s Kaggle dataset, featuring a 4-stage architecture with decoupled ML pipelines for feature processing, training, and real-time inference; implemented a two-tower neural network for customer and item embeddings stored and retrieved via Hopsworks Vector Index.
 - Built end-to-end ML pipelines including feature engineering, retrieval and ranking model training (using CatBoost), item embedding computation, and deployment; integrated Hopsworks Feature Store, Model Registry, and Model Serving, complemented by an interactive Streamlit frontend for recommendations and feedback.

- AnimeAgents — AI Anime Character Simulation Engine**
- Developed an AI agent simulation engine for dynamic anime character interactions using LLMs with Retrieval-Augmented Generation (RAG), LangGraph orchestration, MongoDB memory systems, real-time WebSocket communication, and Groq-accelerated inference.
 - Deployed RESTful APIs with FastAPI and Docker, implemented LLMOps best practices including prompt versioning, automated evaluation with Opik, and utilized Python tooling (uv, ruff) for efficient development and end-to-end system integration.

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| AWARDS | <ul style="list-style-type: none"> • Gold Medal for Academic Excellence (2025) – Awarded by the Alumni Association of the University of Kelaniya. • Gold Medal for Outstanding Performance (2025) – Awarded by the Faculty of Computing and Technology. • Google Cloud Research Credits Program (Award GCP19980904) – Awarded \$1000 in research credits |
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PROFESSIONAL MEMBERSHIPS	Member, Association for Computational Linguistics (ACL)
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| EXTRA CURRICULAR | Huawei Seeds for the Future - 2023 <ul style="list-style-type: none"> • Team Leader for Team Sri Lanka at Huawei Seeds for the Future - 2023 – Selected as one of only four top university students from a highly competitive pool, earning the opportunity to visit China. • Received mentorship from top industry experts on cutting-edge technologies including 5G, AI, Cloud Computing, and Digital Power. • Participated in the Global Tech4Good competition, honing skills in leadership, problem-solving, and entrepreneurship. |
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| ENGLISH PROFICIENCY | IELTS Academic - 2025 <ul style="list-style-type: none"> • Overall Band Score: 8.5 • Listening: 9.0 Reading: 8.5 Writing: 7.5 Speaking: 8.0 |
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REFERENCES

Prof. Sidath R. Liyanage

Professor, Department of Software Engineering

Faculty of Computing and Technology, University of Kelaniya

Email: sidath@kln.ac.lk, **Phone:** 071 649 8614

Dr. Tharindu Cyril Weerasooriya

Research Scientist, Accenture (USA)

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